

Robust Model Predictive Control via Conformal Prediction

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Introduction. Model predictive control (Rawlings et al., 2017) is an established and widely used method in control theory. More specifically, it is a feedback control strategy that uses a dynamic model of the system under control to predict its future behavior over a finite time horizon. At each time step, it solves an optimization problem to determine the best control action, based on the predicted system behavior.

In real applications, the prediction of the future system behavior is commonly afflicted with uncertainty. In this work, we therefore consider the use of conformal prediction (Vovk et al., 2005) to increase safety and make control more robust. Broadly speaking, the idea is to conformalize the predicted system behavior, replacing precise trajectories by confidence bands that cover the true behavior with high probability. Safe control inputs can then be selected in a more “cautious” way by optimizing worst-case scenarios.

Model Predictive Control (MPC). Consider a discrete-time dynamical system $x_{t+1} = f(x_t, u_t)$, where $t \in \mathbb{N}$ is the time index, $x_t \in \mathbb{R}^{n_x}$ is the state, $u_t \in \mathbb{R}^{n_u}$ is the control input, and n_x and n_u denote the dimensions of the state and control spaces, respectively. The transition function is $f : \mathbb{R}^{n_x} \times \mathbb{R}^{n_u} \rightarrow \mathbb{R}^{n_x}$. MPC is a feedback control strategy in which the control action is obtained, at each time step t , by solving a finite-horizon optimal control problem online. Using a model of the system to predict its future evolution, MPC computes an optimal sequence of control inputs $u_{0|t}^*, \dots, u_{N-1|t}^*$ over the prediction horizon $N \in \mathbb{N}$. MPC applies only the first control input of that sequence $u_t = u_{0|t}^*$, and repeats the optimization at the next time step $t + 1$ with updated state information. The ability to handle constraints on states and inputs directly is one of the main reasons for MPC’s wide use in process and motion control (Rawlings et al., 2017). Given the state x_t at time t , MPC solves the optimization problem

$$\min_{u_{0|t}, \dots, u_{N-1|t}} \sum_{i=0}^{N-1} \ell(x_{i|t}, u_{i|t}) + \ell_f(x_{N|t}) \quad \text{s.t.} \quad \begin{cases} x_{i+1|t} = f(x_{i|t}, u_{i|t}), x_{0|t} = x_t \\ x_{i|t} \in \mathcal{X}, u_{i|t} \in \mathcal{U}, x_{N|t} \in \mathcal{X}_f \end{cases},$$

where ℓ and ℓ_f are the stage and terminal cost, $\mathcal{X}$, $\mathcal{U}$ and $\mathcal{X}_f \subseteq \mathcal{X}$ are the state, input and terminal constraint sets. Suitable terminal cost and terminal set are used to guarantee closed-loop stability and recursive feasibility (Mayne et al., 2000).

Although the formulation above is deterministic, uncertainty may arise in MPC from the dynamics, state estimates, constraint parameters or others (Sale et al., 2026). When modeled probabilistically, the predicted trajectory cost becomes a random variable. Stochastic MPC therefore optimizes a functional (e.g. expectation) of this cost (Mesbah, 2016; Kouvaritakis et al., 2016).

Existing Work on MPC with Conformal Prediction. Conformal prediction (CP) is a model-agnostic, distribution-free method for building prediction sets with marginal coverage guarantees (Vovk et al., 2005). In the context of MPC, such prediction sets have been used to tighten constraints in the control problem. Trajectory predictors forecast the motion of surrounding agents, and the resulting uncertainty regions enlarge the safety constraints to avoid collisions with high probability (Lindemann et al., 2023; Dixit et al., 2022; Cleaveland et al., 2024). These approaches have been applied for safe control of a self-driving car at a pedestrian-filled intersection, and in a drone avoiding a moving object. For linear stochastic systems, CP has been used in the system model itself using uncertainty regions of the error trajectory to tighten chance constraints (Vogel et al., 2026).

Our Proposal. Current approaches to stochastic MPC typically address problems such as finding control sequences that minimize the *expected* cost over possible trajectories. In safety-critical applications, one might rather be interested in less risky control, and hence prefer, e.g., control inputs that minimize the expected loss in the upper tail of the cost distribution. This objective can be formalized using the Conditional Value at Risk (CVaR).

We consider a related though slightly different objective based on a quantile of the trajectory-cost distribution. This objective may reduce sensitivity to extreme costs under heavy-tailed uncertainty. In particular, one might be interested in an action sequence minimizing the $1 - \alpha$ quantile of the cost distribution over the horizon N . To this end, we will solve the following minimization task for a user-defined miscoverage level α :

$$\min_{u_{0|t}, \dots, u_{N-1|t}} Q_{1-\alpha} \left(\sum_{i=0}^{N-1} \ell(x_{i|t}, u_{i|t}) + \ell_f(x_{N|t}) \mid x_{0|t}, u_{0|t}, \dots, u_{N-1|t} \right)$$

Our proposal is to construct conformal prediction sets for trajectories and their associated costs. For tackling the exchangeability problem, we are planning to reuse the approach from Stankeviciute et al. (2021). Given a prediction set $C_{1-\alpha}(x_t, u)$ for possible cost assignments induced by a candidate control sequence $u = (u_{0|t}, \dots, u_{N-1|t})$, we can optimize the control sequence with respect to an upper bound $g(u) = \sup C_{1-\alpha}(x_t, u)$ on the corresponding cost. Thus, the optimization procedure yields an optimal control

$$u^* \in \arg \min_{u \in \mathcal{U}^N} g(u).$$

In the full version of the paper, we will develop the technical details on the CP process (specification of inputs, outputs, nonconformity function, etc.) as well as the optimization problem to be solved in each time step. Moreover, we provide theoretical results on formal guarantees as well as empirical results on simulated control benchmarks.

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